
CMP6200/DIG6200
Individual Undergraduate Project
Project Interim Report
Sign Language Translation System

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1.0 Introduction and Context

Sign Languages are natural languages used by millions worldwide; however, communication between sign-language users and non-signers, or individuals with limited proficiency and practice, remain a major accessibility gap. Consequently, this persistent communication gap has motivated significant research into assistive technologies that can bridge the divide through automated recognition and translation of sign language gestures (Ang et al., 2023; Abdullahi and Chamnongthai, 2022).

Recent advances in wearable sensing, particularly surface electromyography (sEMG) combined with inertial measurement units (IMUs), allow for precise tracking of arm and hand gestures in real time. sEMG captures underlying neuromuscular activity, while IMUs track motion patterns, enabling robust gesture recognition (Myo Armband studies, 2015–2022). Complementary use of electroencephalography (EEG) provides high-level cognitive context such as attention and engagement that can enhance the expressiveness of translations without replacing detailed gesture detection.

Building on these technological insights, this project proposes a hybrid approach where a wearable forearm or wrist device collects sEMG and IMU data to identify hand and wrist gestures, while an accompanying EEG module like that of Brain-Life may be integrated to enrich the system with cognitive context such as attention or emotional tone. These signals are processed on a smartphone, where gestures are translated into coherent sentences using sequence-to-sentence models. The final output is delivered as text and/or speech, potentially modulated by EEG-derived emotional indicators that offer a practical and accessible communication aid that supports more natural and expressive interactions.

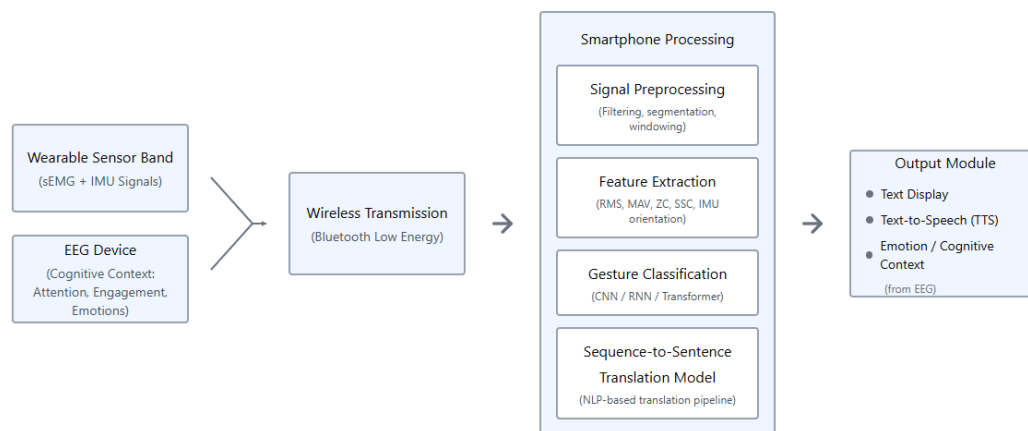


Figure 1. Sign Language Translation System Pipeline with sEMG, IMU & EEG inputs

1.1 Problem Statement

Current sign-language communication solutions are limited by high costs, dependence on human interpreters and low real-time availability. This project addresses the need for a wearable, sensor-based system capable of translating sign language gestures into text or speech in real time according to need, enhancing communication accessibility for individuals with hearing and speech impairments.

1.2 Significance

The significance of this initiative is reflected in several connected ways:

- It enables and enhances daily communication accessibility and thus promotes independence and inclusion, allowing users to interact more freely in social, educational and professional settings.
- By focusing on usability, calibration for different users and adaptable output modes (text and audio), the project aligns with the United Nations' Sustainable

Development Goal 10: Reduced Inequalities, emphasizing accessibility and technological inclusion.

- The system's design leaves room for future enhancements such as emotion-aware speech synthesis, multi-language translation, smartwatch integration for wearable output and cloud-based personalization using federated learning.

2.0 Review of Existing Knowledge

Sign language recognition has been an active area of research for decades with the primary goal of enabling communication between signers and non-signers. Early approaches relied on vision-based systems such as cameras and sensor-gloves to track hand movements and/or while effective for basic gestures, these systems were limited by lighting conditions, occlusions and user dependency. More recently, wearable sensors particularly surface electromyography (sEMG) and inertial measurement units (IMUs) have become central to research, offering higher precision, real-time capabilities and user-independent performance (Gu et al., 2022; Amor et al., 2023).

2.0.1 Wearable Sensor System

sEMG sensors detect the electrical activity produced by muscle contractions, capturing the fine-grained neuromuscular patterns underlying hand and arm gestures. IMUs which measure orientation and motion complement sEMG by providing spatial and temporal information. Systems like the Myo Armband (discontinued) and custom EMG-IMU forearm devices have demonstrated effective gesture recognition for sign language applications (Gu et al., 2022). The strength of these systems lies in sensor fusion, where combining multiple data streams improves classification accuracy and generalization across users.

2.0.2 Machine Learning Approaches

Time-series classification models, including Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks and Transformers, have been widely applied to sEMG and IMU data for gesture recognition. Studies show that deep learning models outperform classical classifiers like SVM or Random Forests in capturing temporal dependencies and complex movement patterns. Preprocessing techniques such as filtering, windowing, rectification, RMS feature extraction and normalization remain critical for noise reduction and robust feature representation.

2.0.3 EEG Augmentation and Cognitive Context

Emerging research explores EEG-based augmentation to capture cognitive states, including attention, engagement and emotional tone. While EEG alone is insufficient for detailed gesture recognition, it provides contextual cues that can enrich communication systems. Integrating EEG with sEMG and IMU may allow translation systems to adapt outputs based on cognitive state, making communication more expressive and natural.

2.0.4 Sentence Level Translation and Mobile Implementation

Most prior work focuses on isolated gesture recognition with limited consideration for sentence-level translation or mobile real-time deployment (Gu et al., 2022; Amor et al., 2023). Sequence-to-sequence models have been proposed to map gesture sequences to coherent sentences, but these are rarely implemented in wearable, portable systems. Real-time translation remains a challenge due to latency, computational constraints and the need for user calibration to account for physiological variability.

2.1 Critical Analysis and Gap Identification

A critical review of current literature reveals that while sEMG and IMU-based systems reliably detect isolated gestures, they are often limited to word-level recognition and offline processing (Gu et al., 2022; Amor et al., 2023). EEG augmentation studies provide insights into cognitive states, yet they rarely integrate these signals into gesture translation for real-time applications.

Existing research also demonstrates high user dependency, with models performing poorly across uncalibrated subjects, highlighting the need for robust cross-user generalization. Moreover, mobile deployment remains largely unexplored; most systems rely on desktop setups or lab-based environments thus limiting accessibility.

These limitations indicate several gaps like:

1. Lack of sentence-level translation from gesture sequences.
2. Insufficient integration of EEG for cognitive context.
3. Limited research on real-time, mobile and user-adaptive implementations.
4. Minimal evaluation of usability, accessibility and user experience in practical settings.

2.2 Project Justification

This project directly addresses these gaps by developing a hybrid wearable system capable of real-time sentence translation, cognitive context adaptation and mobile accessibility for users with hearing or speech impairments. It essentially integrates:

1. Sequence-to-sentence translation, converting gesture sequences into coherent, natural sentences.
2. EEG-based cognitive context, enhancing communication with attention and emotion-aware outputs.
3. Mobile deployment, ensuring accessibility and usability in real-world environments.
4. User adaptability, including calibration and cross-subject generalization to accommodate diverse users.

3.0 Project Aims, Objectives and Scope

3.1 Project Aim

The aim of this project is to develop a wearable system that enables real-time translation of sign language into text and audio to support communication for individuals with hearing and speech impairments.

3.2 Project Objectives

Objective 1: Mobile App Design and Setup

(Specific) Design and build the base smartphone application keeping Nielsen's Usability Heuristics in mind that will later handle gesture input, sentence display and text-to-speech (TTS) output.

(Measurable) Completion of a functional UI layout including a connection-setup screen, calibration screen and live translation screen.

(Achievable) Use React Native/Flutter with built-in TTS (Text-to-Speech) and accessibility options.

(Relevant) A functional mobile interface is essential for real-time translation and accessibility by end-users.

(Time-limited) To be delivered by the conclusion of Week 2.

Objective 2: Wearable Device Integration

(Specific) Establish Bluetooth Low Energy (BLE) communication with a multi-channel sEMG & IMU wearable (e.g. Custom forearm band like that crafted by Rombolab, OpenEMG, FREEEMG, Myo or similar) and pair the BrainLife EEG module for cognitive-context data.

(Measurable) A stable Bluetooth data stream with live visualization of raw sEMG , IMU and EEG signals to work on.

(Achievable) Use existing BLE libraries like React Native BLE PLX, CoreBluetooth or Android BLE for stable communication.

(Relevant) Reliable sensor communication is foundational for accurate signal acquisition and gesture recognition.

(Time-limited) Fully integrated and tested by Week 6.

Objective 3: Dataset Construction and Signal Preprocessing

(Specific) Develop a training dataset that integrates publicly available EMG/IMU datasets (such as Ninapro, Myo-based datasets, EMG-Healthy, etc.) with limited custom data gathered via the application for personalized calibration.

(Measurable) A unified dataset stored in CSV or NumPy format, labelled with gesture names, timestamps and metadata, cognitive EEG state tags.

(Achievable) Use signal preprocessing techniques like filtering, rectification, windowing, FFT features and normalization.

(Relevant) A clean, well-structured dataset is crucial for training accurate machine learning models.

(Time-limited) Finish dataset integration, cleaning and preprocessing by Week 10.

Objective 4: Gesture Recognition Model

(Specific) Train and evaluate machine learning models to recognize gestures from EMG and IMU signals. Aim for at least 85% accuracy on data.

(Measurable) Performance metrics including accuracy, confusion matrices and F1-scores.

(Achievable) Use both classical models (e.g., SVM) and deep learning architectures (e.g., CNN, RNN or Transformer) for comparison.

(Relevant) High-quality gesture classification is the core of the translation pipeline and determines system usability.

(Time-limited) Complete and validate gesture recognition models by Week 14.

Objective 5: Sentence Translation and App Integration

(Specific) Develop a sentence translation layer that converts recognized gesture sequences into natural sentences. Use a small paired dataset or synthetic examples linking gesture-word sequences to grammatically correct sentences (e.g., “HELLO HOW YOU” → “Hello, how are you?”).

(Measurable) Evaluate translation quality using BLEU scores and human review for clarity and fluency.

(Achievable) Begin with a rule-based approach for limited phrases and later expand to a lightweight sequence-to-sequence (seq2seq) transformer model. Integrate the final translation output into the app for text and TTS display.

(Relevant) Users require meaningful sentences, not isolated gesture labels; hence translation enables real communication.

(Time-limited) Complete model integration and functional testing by Week 18.

Objective 6: User Testing and Evaluation

(Specific) Conduct small-scale user testing with 8-10 participants to assess usability, translation accuracy, response speed in real-world use and user comfort.

(Measurable) Gather feedback using structured surveys and measure accuracy, latency and ease of use.

(Achievable) Recruit classmates or volunteers under informed consent and standard ethical guidelines.

(Relevant) Testing ensures the system meets real-world usability, accessibility and performance needs.

(Time-limited) Complete all testing and evaluation by Week 20.

3.3 Project Scope

Listed below is the scope of the project:

Category	Details
In-Scope	1. Development of a prototype mobile application (UI screens: connection setup, calibration, live translation, text display + TTS). 2. Integration of wearable hardware via BLE (sEMG + IMU band with an optional EEG module) with real-time data streaming and visualization. 3. Dataset construction using public EMG/IMU datasets + limited custom calibration data; preprocessing (filtering, windowing, normalization, feature extraction). 4. Machine learning model development for gesture recognition (classical + deep learning) and a lightweight sentence translation layer. 5. Full pipeline integration: gesture input => ML prediction => sentence mapping => app display/TTS. 6. User testing with 8-10 participants for usability, accuracy and latency evaluation.
Out-of-Scope	1. Full sign language translation covering entire vocabularies or grammatical structures. 2. Custom fabrication of advanced hardware or medical-grade sEMG devices. 3. Large-scale user trials, clinical validation, or statistically powered studies. 4. Advanced NLP for natural, unrestricted sign-to-

	speech generation beyond limited phrase mapping. 5. Cross-platform deployment to app stores or long-term product maintenance/support.
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Table 1. Scope of the project

While several elements remain out of scope for this project, this does not mean they will never be attempted or completed.

4.0 Project Design

4.1 Design and Methods

This project adopts a development-focused, experimental, user-centred design approach combining system development, sensor-data experimentation and small-scale user evaluation involving the following methodological components:

1. System Development

- **Mobile App Development:** Using React Native/Flutter to build the UI, BLE connectivity layer and TTS output.
- **Wearable Device Integration:** Implementing BLE streaming from sEMG, IMU and EEG modules into the app.
- **Real-time Processing Pipeline:** Designing a pipeline for signal calibration, filtering, feature extraction, classification and sentence translation.

2. Experimental Data Processing

- **Signal Acquisition:** Collecting EMG, IMU and EEG data from public datasets and small-scale in-house recordings.
- **Preprocessing & Feature Engineering:** Filtering, rectification, segmentation, FFT features, RMS/MAV and standardisation.
- **Model Training:** Using classical ML (SVM, RF) and deep learning models (CNN, RNN, Transformer) for gesture classification.
- **Sentence-Level Translation:** Implementing a rule-based baseline and a lightweight seq2seq model for gesture-to-sentence generation.

3. Usability & Evaluation

- **User Testing:** 8-10 participants will test the prototype for gesture recognition accuracy, latency and usability.
- **Evaluation Methods:**
 - **Quantitative:** accuracy, F1-score, BLEU scores, latency measurements
 - **Qualitative:** surveys, interviews, user feedback

4.2 Justification of Methods

The techniques chosen for this project were based on their technical suitability, feasibility and alignment with the project's objectives.

A multimodal wearable-sensor strategy that integrates sEMG, IMU and EEG was chosen because it allows for a more comprehensive representation of hand gestures and user context since sEMG captures subtle muscle activations, IMU supplies movement and orientation information and EEG provides cognitive or emotional signals that can enhance expressive output. This approach directly addresses the shortcomings of previous studies that primarily concentrated on isolated, offline EMG gesture classification without considering sentence-level translation or emotional subtleties.

As for the development of the model, deep learning frameworks such as CNNs, LSTMs and Transformers are utilized due to their established success in analysing multichannel bio signals and their capacity to learn intricate patterns without the need for extensive manual feature engineering; traditional models like SVMs and Random Forests are also incorporated as benchmarks to ensure a thorough comparison. A lightweight seq2seq translation method is implemented to advance beyond basic gesture recognition and introduce natural language structure, addressing a significant gap in the literature where gesture-to-sentence generation is seldom explored.

The project also employs a user-centred evaluation approach, utilizing small-scale participant testing to evaluate usability, latency and recognition performance. Alternative techniques such as vision-based recognition, glove-based sensors and large EEG sentiment models were deliberately omitted due to concerns regarding privacy, intrusiveness, high hardware costs or excessive complexity.

4.3 Project Timeline

The development of this project can be divided into a 20-week plan consisting of the following key tasks and their respective sub-tasks (omitting the time frame allotted for proposal and report writing at the very beginning to receive necessary feedback to commence). The bullet points below and the Gantt chart illustrates the breakdown and dependencies of each activity:

Task 1 – Application Design and Setup

1.1. UI/UX Design: Create simple, accessible screens for device connection, calibration and displaying translation outputs.

1.2. TTS Integration: Implement basic text-to-speech functionality to convert translated text into audio for accessibility.

Press the following link to view if chart is unclear: [HTA flowchart](#)

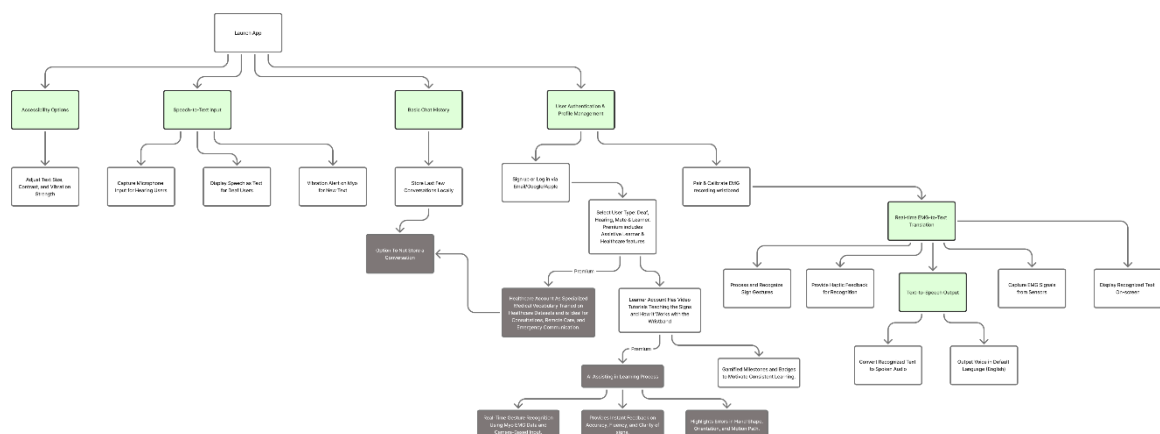


Figure 2. HTA flowchart of the application

Task 2 – Wearable Device Integration and Calibration

- 2.1. Device Setup and BLE Device Connection: Establish a reliable Bluetooth Low Energy (BLE) connection between the wearables and the smartphone app.
- 2.2. Signal Calibration: Calibrate EMG, IMU and EEG signals for user-specific sensitivity and noise reduction to improve recognition accuracy.

Task 3 – Dataset Preparation

- 3.1. Public Dataset Review: Examine publicly available EMG datasets such as Myo-Gesture, Ninapro and compatible EEG datasets for baseline modelling.
- 3.2. Custom Data Collection: Record additional gesture data from participants to expand and personalize the dataset.
- 3.3. Data Cleaning and Labelling: Merge, standardize and label data samples to ensure consistent structure and model-ready inputs.

Task 4 – Model Development

- 4.1. Feature Extraction: Transform raw EMG, IMU and EEG readings into meaningful features (e.g., RMS, MAV, IMU orientation vectors etc.) for classification.
- 4.2. Model Training: Train gesture-recognition models using classical algorithms (SVM, Random Forest) and deep learning architectures (CNNs, RNNs, Transformers).
- 4.3. Model Validation: Evaluate performance using metrics such as accuracy, precision, recall and F1-score to verify reliability.

Task 5 – Translation and App Integration

- 5.1. Paired Dataset Creation: Construct a paired dataset mapping recognized gestures to sentence-level outputs.
- 5.2. Translation Model: Implement and train a lightweight sequence-to-sequence (seq2seq) model or rule-based translator to convert gesture tokens into

sentences.

5.3 App Integration: Integration translation results, TTS output and optionally EEG-informed emotional modulation to create more expressive outputs.

Task 6 – Testing and Evaluation

6.1. Participant Testing: Recruit 8-10 participants to test the prototype for gesture accuracy and ease of use.

6.2. Feedback Collection: Gather user feedback through surveys and interviews on usability, latency and accessibility.

6.3. Final Evaluation: Analyse testing data, summarize results and identify improvements for future development.

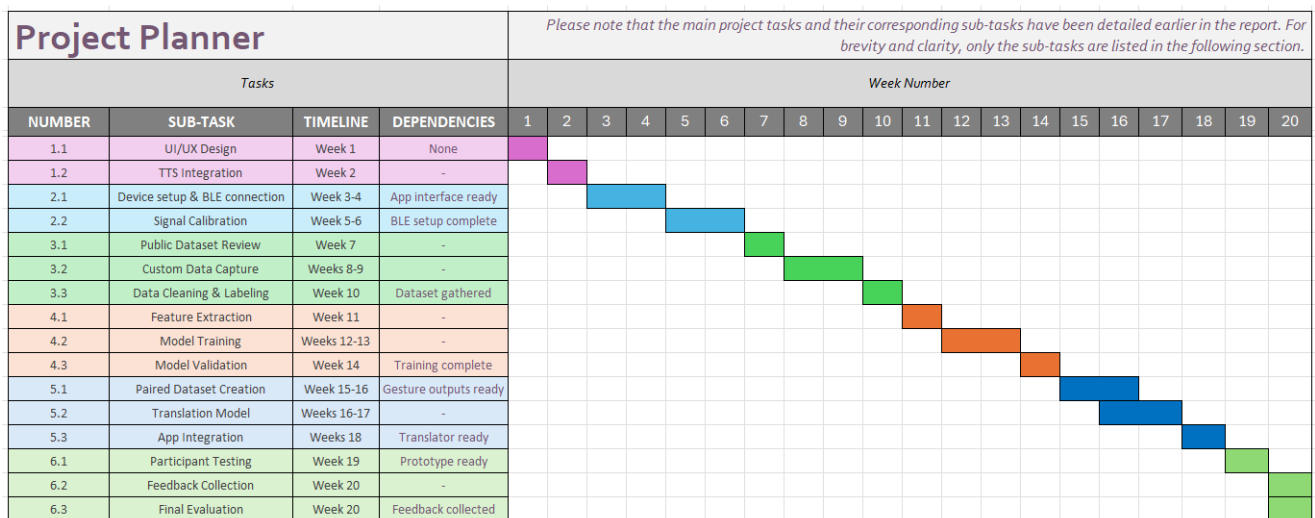


Figure 3. Gantt chart of the project timeline

5.0 Feasibility, Risks and Ethical Aspects

5.1 Feasibility

This project is feasible. The scope has been deliberately crafted around methods, tools and hardware that are realistically accessible like consumer-grade sEMG/IMU

wearables, supervisor-provided EEG modules and open-source machine learning libraries like TensorFlow and PyTorch.

The primary development tasks (Bluetooth integration, data preprocessing, model training and app development) correspond with the technical skills that have already been developed through prior coursework and hands-on experience in mobile development, Python-based machine learning and sensor interfacing.

Publicly accessible datasets like Ninapro and Myo-based EMG sets allow for model training to begin even before the custom data collection is finalized, which minimizes reliance on participant availability.

The structured 20-week timeline also demonstrates that tasks are appropriately distributed with buffer periods to accommodate unexpected delays.

5.2 Risk Analysis and Mitigation

Potential Risks	Likelihood	Impact	Risk and Mitigation Strategy
Hardware availability	Medium	High	Delays in assembling the custom sEMG–IMU wearable or acquiring a substitute may cause significant delays. The only way to deal with this issue is to begin app and model setup until the hardware is available.
Bluetooth connection issues	Medium	Medium	BLE communication between the wearable device and smartphone may be unstable or experience packet loss. We can deal with this issue by conducting early connection

			testing and use wired data logging as a fallback during development.
Poor Signal Quality	Medium	High	Noise, electrode displacement, or inconsistent contact can degrade model accuracy. This can be mitigated by enforcing calibration routines, using filtering & signal quality checks and apply standardized electrode placement.
Insufficient dataset volume	Medium	High	Limited data may reduce model accuracy. This can be dealt with by combining public datasets with user-collected data and apply data augmentation to increase robustness.
Model overfitting or poor accuracy	Low	High	Model may perform well on training data but poorly on new users. We can deal with this by using cross-validation, dropout and gathering calibration samples from additional users if possible.
Time constraints	Medium	High	Tight timeline may cause testing or reporting delays. The best way to deal with this is to assign a few extra weeks which will be possible as the Gantt chart shows that as per planning only 20 weeks will be used and we have more time as per allotment.

Limited participant availability	Medium	Medium	Difficulty in recruiting 8-10 volunteers for testing. This can be prevented by beginning recruitment early.
Ethical or privacy concerns	Low	High	Risk of collecting personal data without consent. This won't be an issue if we obtain written consent, anonymize datasets and comply with university ethics policy.
Integration and latency issues	Medium	Medium	Gesture-to-speech output may have delays. To deal with it we can optimize inference and use lightweight models to ensure faster response time.
Software or library incompatibility	Low	Medium	Issues may arise due to updates in TensorFlow, PyTorch, React Native, or BLE libraries. We could just freeze versions using requirements.txt and maintain backups.
Personal issues	Low	Medium	Illnesses or emergencies would very much affect the progress of the project and this is inevitable, but it could be dealt with by communicating promptly with the supervisor and adjusting timelines accordingly.

Table 2. Risk analysis and mitigation

5.3 Ethical Considerations

This project involves the collection of physiological signals and limited user participation and so to maintain ethical responsibility, participants will provide informed consent prior to all data collection with clear explanations regarding the purpose of the project, the nature of the sensors used, the type of data recorded and their right to withdraw at any time. Data will be anonymised using participant codes, ensuring no personally identifiable information is stored or shared. Sensor data will be stored securely in encrypted formats and used strictly for research purposes in accordance with university ethical guidelines. Additionally, the mobile app will not store audio and text outputs at the time of testing.

This project aims to create a positive impact in the field by improving accessibility for individuals both with and without impairments; therefore, all data, resources and outcomes will be used solely for ethical, non-exploitative research purposes with no ulterior intent.

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Appendices

- Rombolabs Project Overview: <https://rombolabs.github.io/#project>